Handout - *Bas van Fraassen, The Scientific Image, Introduction + Ch. 2*

The basic problem

van Fraassen frames the central issue as a conflict between **empiricism** and **scientific realism**. The problem is not whether science is *successful*. Both sides grant that it is. The problem is: **what, exactly, does that success entitle us to believe?** Does the success of science justify belief that our best theories are *true*, including in what they say about **unobservable entities and processes**? Or does it justify only the weaker claim that theories get the **observable world** right? (Intro, pp. 1–5)

van Fraassen begins with the old dispute between those who demand hidden causes or natures behind phenomena and those who resist metaphysical inflation. Early modern mechanists rejected Aristotelian “substantial forms,” but then faced the same pressure at the atomic level: if atoms explain the phenomena, what explains the atoms? (pp. 1–2). This is the recurring temptation of metaphysics: the demand that every regularity have a deeper explanatory ground.

van Fraassen’s proposed solution

van Fraassen’s solution is **constructive empiricism**. Science does not aim at a literally true story about all of reality, including the unobservable. Rather, **science aims at empirically adequate theories**, and **accepting** a theory involves belief only that it is empirically adequate, that is, that it “saves the phenomena” (Intro, p. 4; Ch. 2, pp. 11–13).

This is a carefully limited anti-realism:

- It is **not positivism** or a verificationist theory of meaning.
- It insists that scientific language is to be taken **literally**.
- It denies only that theory acceptance must involve belief in the truth of the theory *as a whole*, especially regarding the unobservable (Ch. 2, pp. 10–12).

van Fraassen’s main claims are:

> **Scientific realism:** science aims at truth, and acceptance of a theory involves belief that it is true. (Ch. 2, pp. 8–9)
> **Constructive empiricism:** science aims at empirical adequacy, and acceptance of a theory involves belief only that which it is empirically adequate. (Ch. 2, p. 12)

Introduction: setting up the dispute (pp. 1–5)

van Fraassen's introduction does two important things.

First, it distinguishes two dimensions of philosophy of science:

- the **structure/content** of theories
- the **relation** of theories to the world and to theory-users (p. 2)

Second, it says that modern scientific theories typically describe both:

- **observable phenomena**
- **unobservable processes/structures** invoked to explain them (p. 3)

The realist conclusion is that science seeks true theories about both. van Fraassen rejects that move. From the empiricist point of view, theories need only be true **about what is observable**; the rest may function as representational machinery or model-building apparatus (p. 3).

A very important feature of the introduction is van Fraassen's distinction between **belief** and **acceptance**. Acceptance is not merely propositional belief. It also involves a **pragmatic commitment**: adopting a conceptual framework, joining a research programme, and using a theory's resources to continue inquiry (p. 4). This allows him to explain how scientists can fully *accept* a theory without thereby believing it to be literally true in every respect.

Chapter 2, §1: realism and its alternative (pp. 6–13)

van Fraassen gives a deliberately minimal formulation of scientific realism:

- **Science aims to give us a *literally* true story of what the world is like.**
- **Acceptance of a scientific theory involves belief that it is true.** (p. 8)

He wants to reject views on which theories are useful myths, metaphors, calculational devices, or merely instrumentally interpreted (pp. 9–10). Realists and constructive empiricists alike take theories ***literally***.

That point is easy to miss. Constructive empiricism is **not** saying:

- electrons are just metaphors
- theoretical discourse is meaningless
- science is mere calculation

Instead, it says: when a theory says “there are electrons,” it does indeed say that there are electrons. The disagreement is not semantic (i.e. does “electron” mean electron) but **epistemic** (i.e. must acceptance of the theory include belief that electrons really exist?). (pp. 11–12)

van Fraassen’s answer is no. Acceptance involves belief only that the theory is **empirically adequate**; that what it says about observable things and events is true. (p. 12)

Chapter 2, §2: the observable/unobservable distinction (pp. 13–19)

One of the realist attacks is that the distinction between observable and unobservable cannot be made, or is philosophically insignificant.

van Fraassen concedes part of the attack:

- language is thoroughly **theory-laden**
- the positivist hope for a purely observation language fails (pp. 14–15)

But this does **not** undermine his position, because constructive empiricism does not depend on a theory/observation vocabulary split. It depends on a distinction between **what is observable** and **what is not**.

His most famous move here is to defend observability as a legitimate, if somewhat vague, notion. Something is observable if there are circumstances such that, were it present under those circumstances, **we would observe it** (p. 16). Vagueness is not fatal. Many legitimate predicates are vague.

The key contrast is:

- **Jupiter’s moons** seen by telescope count as observable, because in principle humans could see them unaided from close enough (p. 16).
- **Particles in a cloud chamber** are not literally observed; what is observed is the vapor trail, from which the particle is **detected** (p. 17).

This section is central because it shows van Fraassen’s **anthropocentric** notion of observability. Observable means observable **to us**, given our epistemic situation as human beings (pp. 17–18). Realists object that this makes ontology depend on human limitations; van Fraassen replies that observability is not meant to determine what exists, but rather to determine the *proper epistemic attitude toward theories* (p. 18).

Chapter 2, §3: inference to the best explanation (pp. 19–23)

Realists often argue that ordinary and scientific reasoning both use **inference to the best explanation** (IBE), so if we infer mice from scratching in the walls, we should also infer electrons from explanatory success. van Fraassen's reply is twofold.

First, the mouse case does not discriminate between realism and empiricism, because the mouse is an **observable entity**. "There is a mouse" and "the observable phenomena are as if there is a mouse" come to the same thing in such a case (p. 21).

Second, even if IBE is legitimate, it only works **within a set of rival hypotheses**. The realist chooses among explanatory hypotheses that include truth about unobservables. The empiricist chooses among hypotheses of the form: *this theory is empirically adequate* (pp. 21–22). So IBE cannot by itself force realism.

Chapter 2, §§4–6: limits of the demand for explanation (pp. 23–34)

Here van Fraassen attacks the intuition that unexplained regularities are intolerable.

Realists like Smart say that unless a theory is true about unobservables, the success of science would be a **cosmic coincidence** or miracle (pp. 24–25). van Fraassen replies that this simply reasserts, rather than proves, the supremacy of explanation.

He sharpens the issue through **Reichenbach's Principle of the Common Cause**: correlations should be explained by common causes (pp. 25–28). But if that principle is made strict and universal, it pushes us toward **hidden variables** and clashes with modern physics, especially quantum theory (pp. 28–31).

So the lesson is: science does value explanation, but not explanation at **any cost**. Explanation is a methodological virtue, not a categorical imperative.

His thought experiment about gold dissolving at different rates makes the same point. A hidden microstructure may be scientifically useful, but what justifies it is not some absolute duty to posit deeper reality. What justifies it is that it may generate new empirical regularities and fruitful theory development (pp. 32–34).

Chapter 2, §7: the "no miracles" argument (pp. 34–40)

Putnam's celebrated line is that realism is "the only philosophy that doesn't make the success of science a miracle" (p. 39). van Fraassen's response is famous and elegant: **the success of science is not miraculous because unsuccessful theories do not survive** (p. 40).

This is his **Darwinian reply**:

- scientific theories are born into competition
- only the successful ones are retained
- therefore their success is not surprising

This does not prove realism. It explains success without saying that accepted theories are true about the unobservable.